PROJECT DAYS · LEICA GEOSYSTEMS

Build cool stuff with real machines

Student projects and a student job at Leica Geosystems Machine Control, Odense.

- Real-Time Mapping of the Construction Site Surface
- Real-Time Pose Estimation for Construction Machines
- Positioning Heavy Machines with IMU + GNSS
- Recognizing Excavator Buckets with a Camera
- Detecting People and Obstacles with Radar + Camera
- Operating Construction Machines Remotely in VR
- **We're hiring: Student Programmer**

For more info, scan here

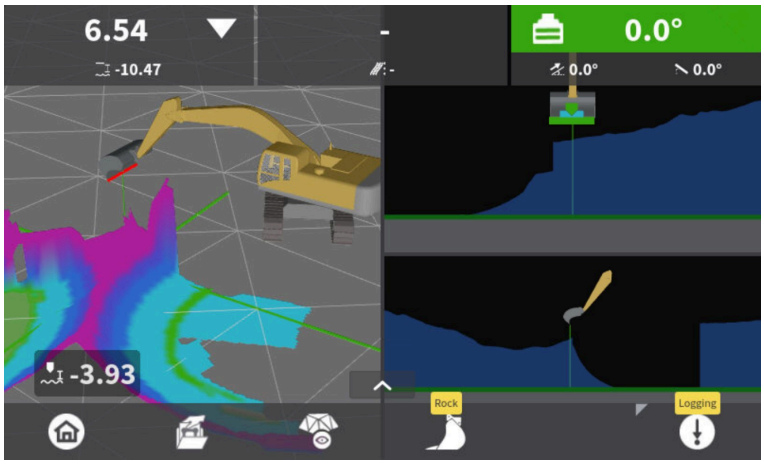
https://sducollab.duckdns.org/

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STUDENT PROJECT · PROJECT DAYS

Real-Time Mapping of the Construction Site Surface

Put cameras around an excavator and turn what they see into a live 3D map of the ground. While the machine digs, the operator always sees the current surface: what has been dug, what is left, and what has changed.



For more info, scan here

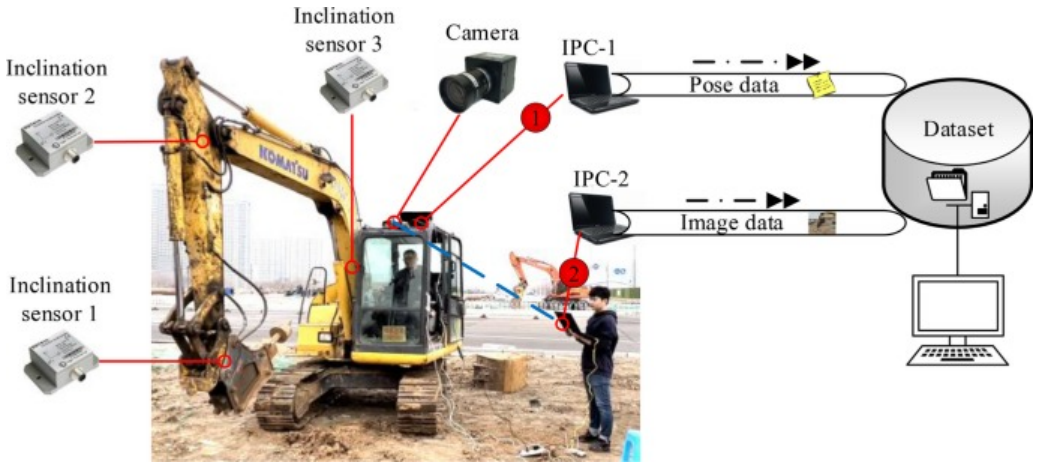
https://sducollab.duckdns.org/#projects/surface-mapping

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STUDENT PROJECT · PROJECT DAYS

Real-Time Pose Estimation for Construction Machines

Use a camera to see where the excavator's arm or the dozer's blade is and how it is angled, live. Combined with inclination sensors and GNSS, the pose stays reliable even when the view is blocked.



For more info, scan here

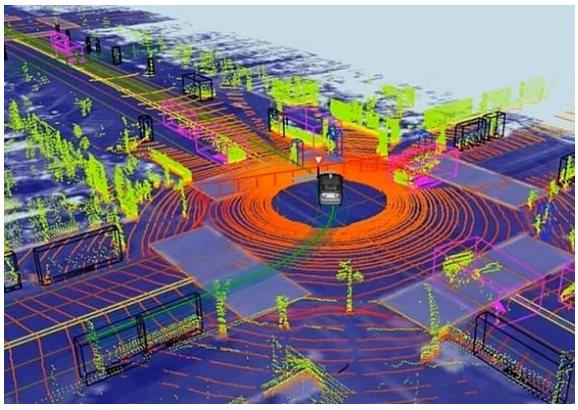
https://sducollab.duckdns.org/#projects/blade-pose

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Positioning Heavy Machines with IMU + GNSS

GPS tells a machine where it is, but drops out near buildings or in deep pits. A motion sensor (IMU) is fast but drifts. Combine the two so an excavator always knows its position and heading, tested on a simulated construction site.



For more info, scan here

https://sducollab.duckdns.org/#projects/ins-sensor-fusion

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Recognizing Excavator Buckets with a Camera

Excavators swap buckets all the time. Train a camera to recognize which bucket is attached, so the machine picks the right settings automatically.



For more info, scan here

https://sducollab.duckdns.org/#projects/tool-recognition

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Detecting People and Obstacles with Radar + Camera

A camera sees what an object is; a radar sees how far away it is, even through dust and fog. Fuse the two so an excavator can warn its operator about people and obstacles around it.



For more info, scan here

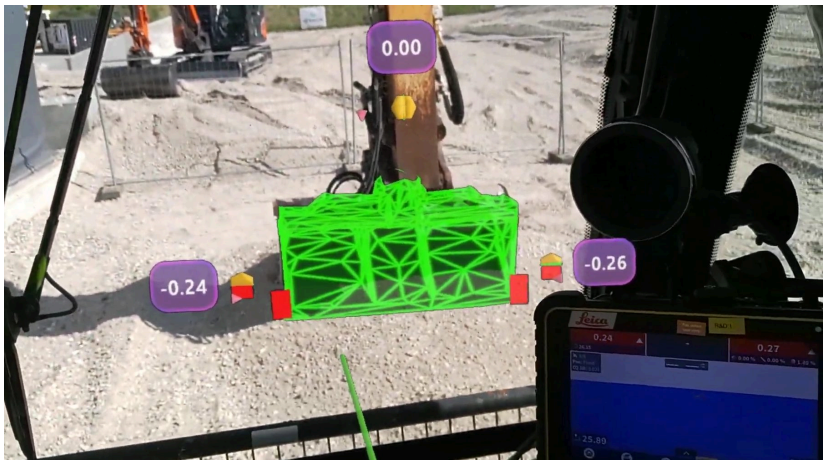
https://sducollab.duckdns.org/#projects/radar-camera

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STUDENT PROJECT · PROJECT DAYS

Operating Construction Machines Remotely in VR

Put on a VR headset and step into a live 3D copy of an excavator and its worksite, built from real survey data. As a final step: drive it remotely from a safe place.



For more info, scan here

https://sducollab.duckdns.org/#projects/vr-teleoperation

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STUDENT JOB · MACHINE CONTROL

We're hiring: Student Programmer

Join our On Machine Software team in Odense. Write, test and improve the software that runs on real excavators and dozers, with access to our T-Site test area.



For more info, scan here

https://sducollab.duckdns.org/#job

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